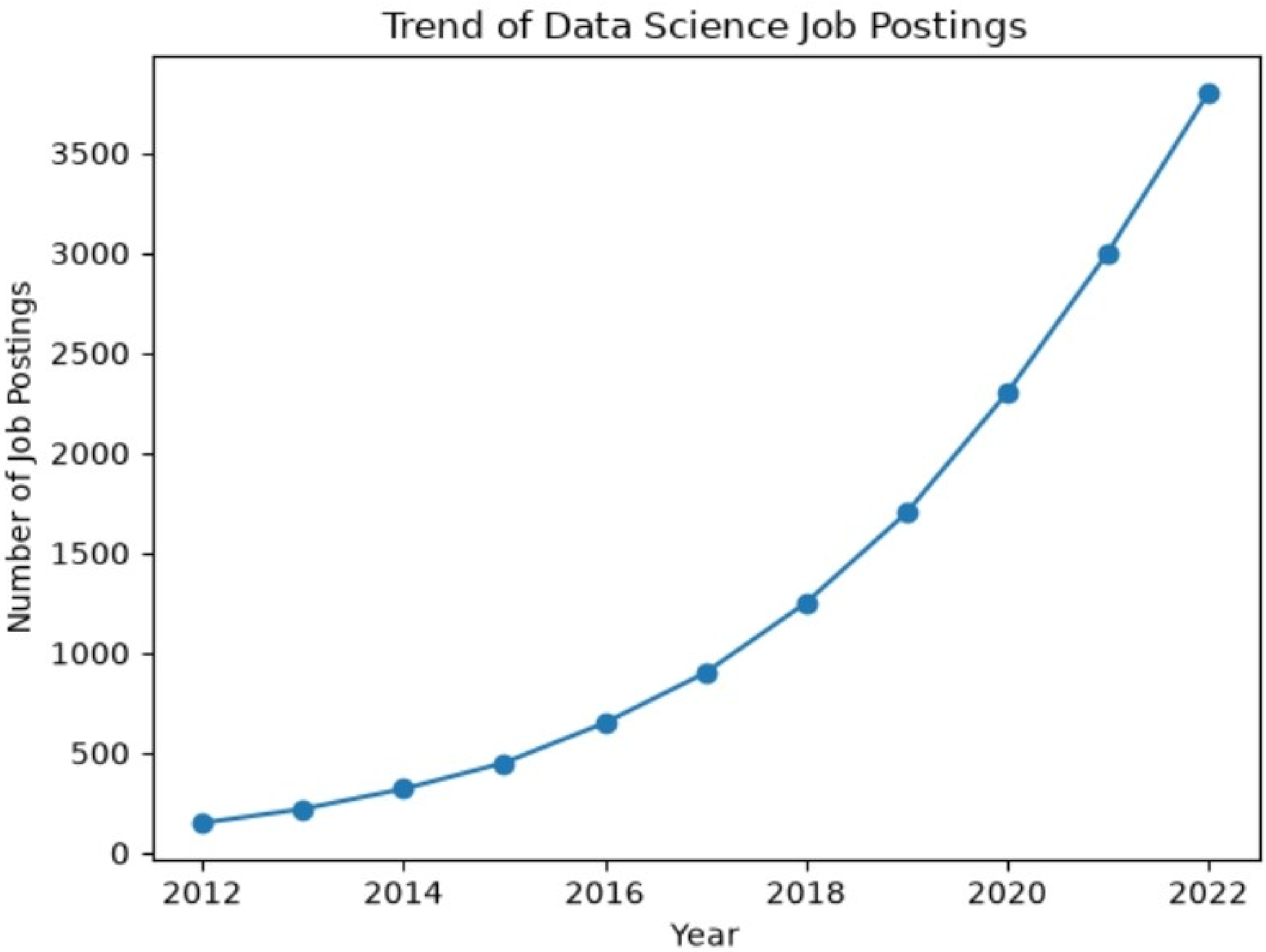
Exp No:1.a Analyze the trend of data science job postings over the last decade

Program:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {

    'Year': list(range(2012, 2023)),

    'Job Postings': [150, 220, 320, 450, 650, 900, 1250, 1700, 2300, 3000, 3800]

}

df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

plt.title("Trend of Data Science Job Postings")

plt.xlabel("Year")

plt.ylabel("Number of Job Postings")

plt.show()
```

OUTPUT:

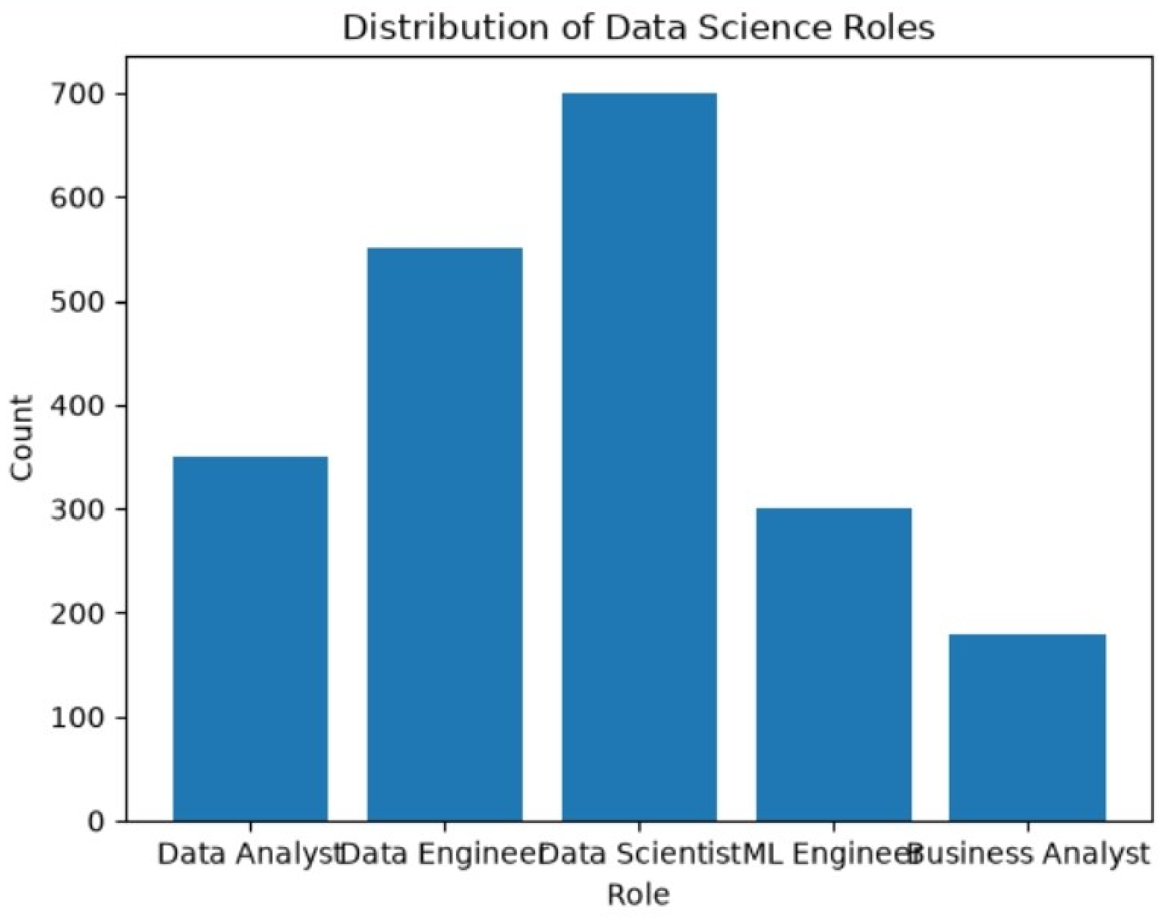


Exp No:1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

PROGRAM:

```
import matplotlib.pyplot as plt

roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer', 'Business Analyst']

counts = [350, 550, 700, 300, 180]

plt.bar(roles, counts)

plt.title('Distribution of Data Science Roles')

plt.xlabel('Role')

plt.ylabel('Count')

plt.show()
```

OUTPUT:



Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi-structured data based on data sets given.

PROGRAM:

```
import pandas as pd

structured_data = pd.DataFrame({

  'ID': [201, 202, 203],

  'Name': ['Alice', 'Bob', 'Charlie'],

  'Age': [24, 27, 30]

})

print("Structured Data:\n", structured_data)

unstructured_data = """This is another example of unstructured data.

It may contain text, images, audio, or video files."""

print("\nUnstructured Data:\n", unstructured_data)

semi_structured_data = {

  'ID': 201,

  'Name': 'Alice',

  'Attributes': {

    'Height': 165,

    'Weight': 60

  }

}

print("\nSemi-structured Data:\n", semi_structured_data)
```

Output

```
Structured Data:

  ID Name Age

0 201 Alice 24

1 202 Bob 27

2 203 Charlie 30

Unstructured Data:

 This is another example of unstructured data.

It may contain text, images, audio, or video files.
```

Semi-structured Data:

{'ID': 201, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 60}}

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

PROGRAM:

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()

f = Fernet(key)

token = f.encrypt(b"Computer Science Department")

print("Encrypted Data:", token)

decrypted_text = f.decrypt(token)

print("Decrypted Data:", decrypted_text)
```

Output:

Encrypted Data: b'gAAAAABqlCsrZwy1gpUFi_Dvi07j8sridmyfyJL-xilubxfyFISW0yKRqLBVHHq2R5-FWsXy2rs0TohbOvwWhRGJzasrDSUM-W9sJLNnppsVgg3s_wSvDyY='

Decrypted Data: b'Computer Science Department'